

Weather risk is more than the weather

Recorded storm deaths per person have fallen. Heat, migration, and the cost of protection make the story more complicated.

68%

Lower recorded storm fatality intensity, 1940-49 to 2010-19.*

438 / year

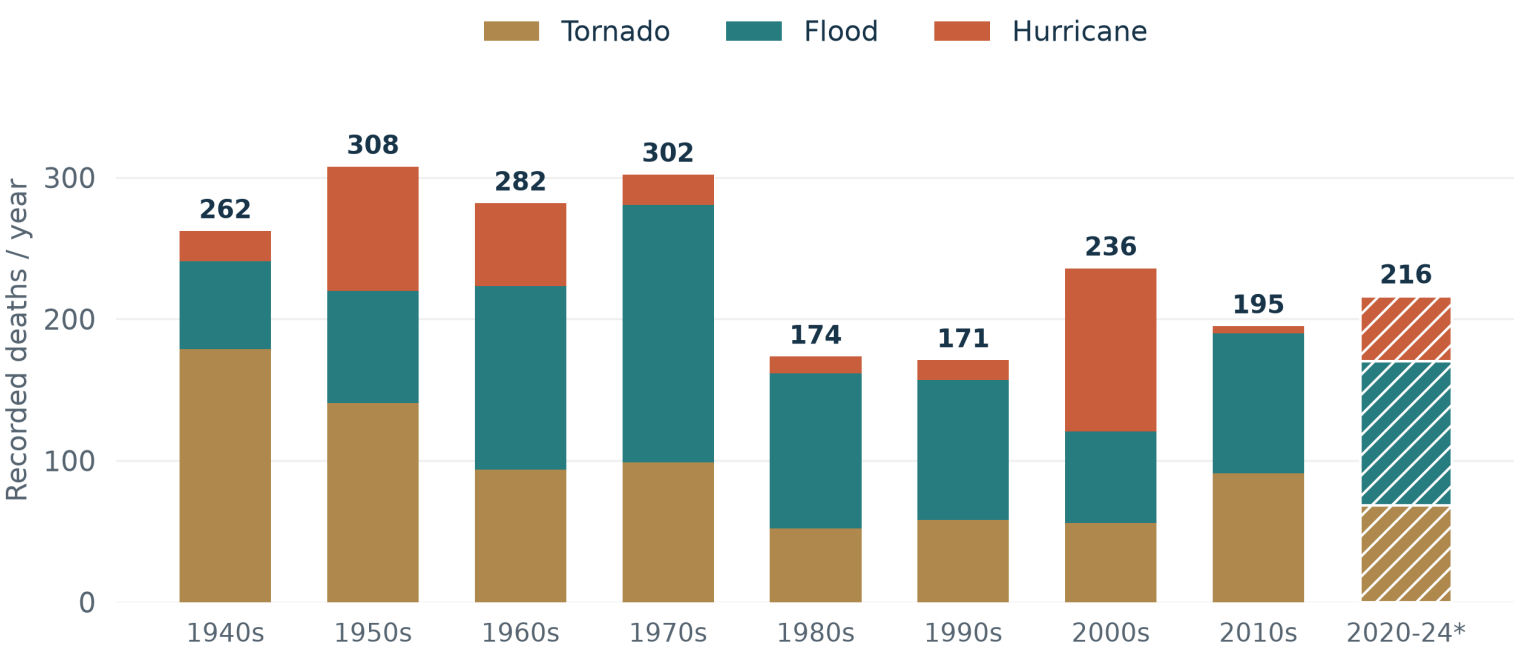
Recorded heat fatalities, 2020-24; 115 per year in 2010-19.

+21,267

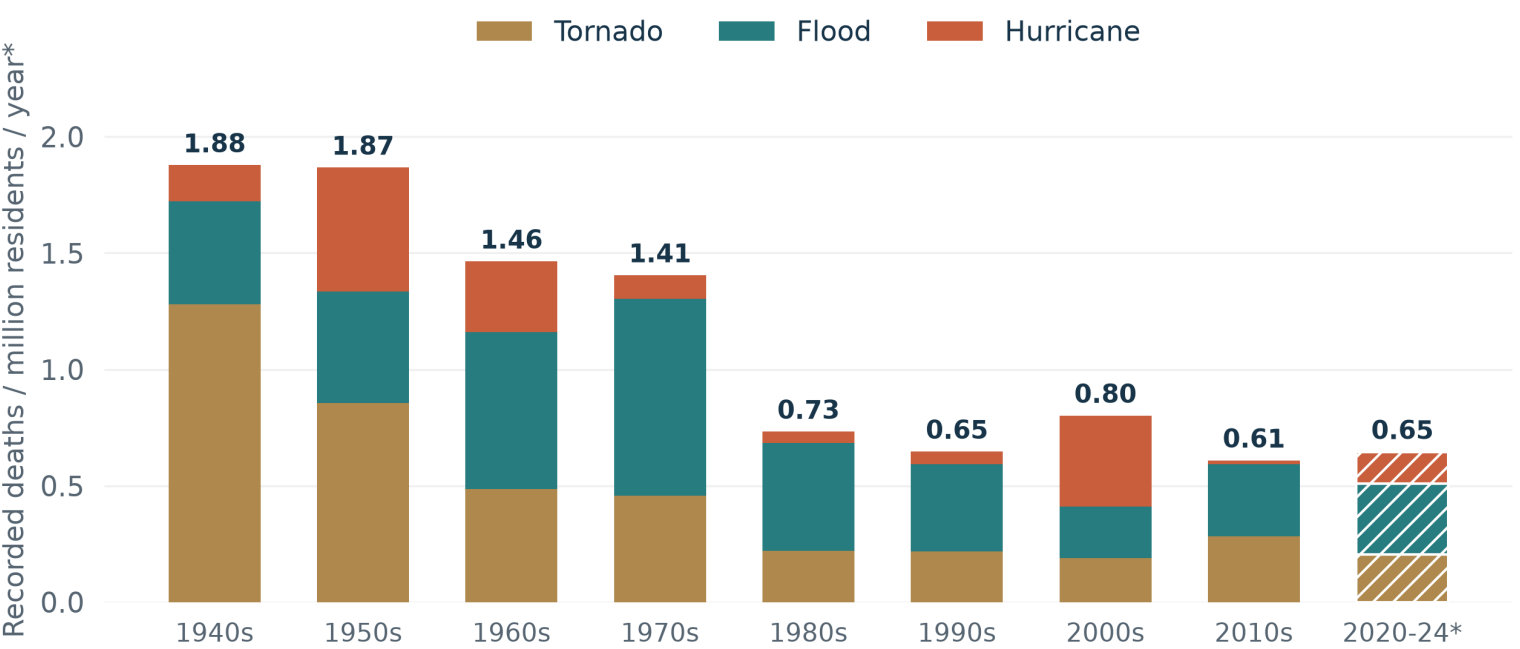
Net domestic migrants to Phoenix metro, July 2023-July 2024.

01 THE DENOMINATOR MATTERS

Annual average fatalities



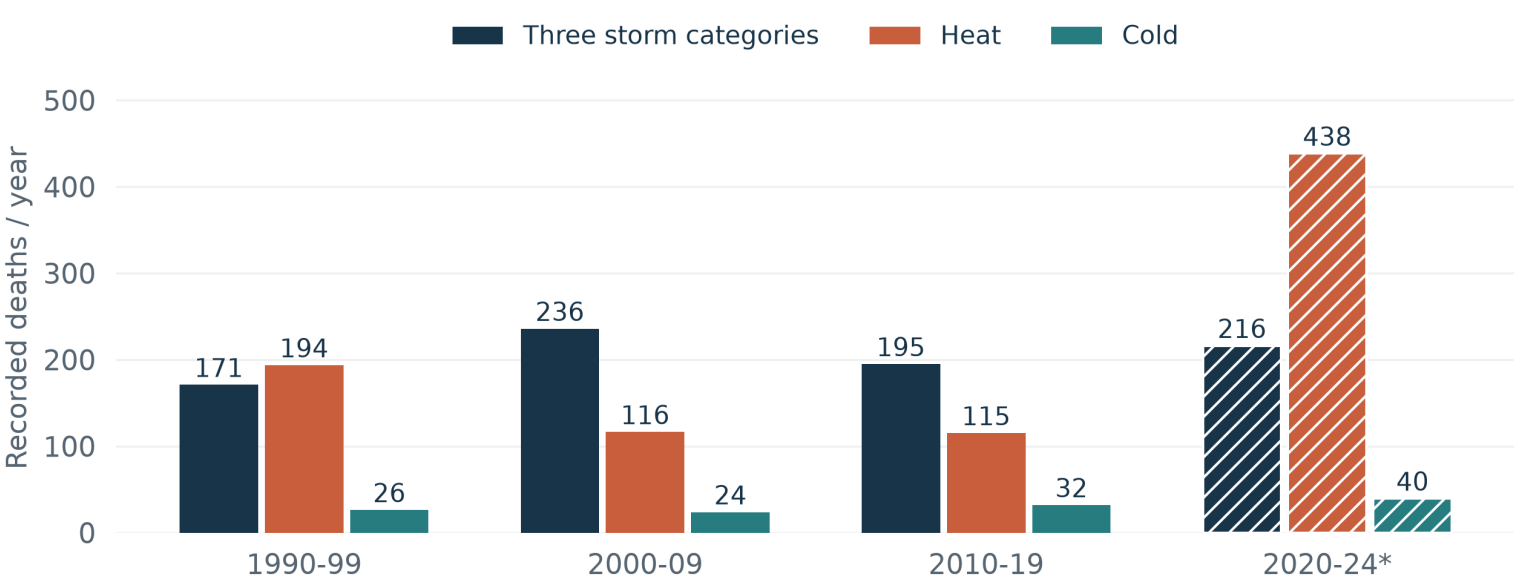
Population-adjusted intensity*



Tornado + flood + hurricane categories. Labels show totals. Full-period average (1940-2024): 239.8 deaths/year.

*NWS reporting includes territories; population covers 50 states + DC. This is an approximate indicator, not a precisely matched mortality rate. Hatched bars show five years (2020-24); other bars show ten. Reporting changes also matter.

02 HEAT CHANGES THE PICTURE



Heat and cold records begin in 1986 and 1988. Missing years are not zero. Cold excludes the separate winter-weather category.

COUNTING DIFFERENT THINGS

NWS event counts record classified weather fatalities. Burke et al. model deaths attributable to temperatures above or below a mortality-minimizing benchmark, including effects on ordinary days and deaths recorded under other causes.

These are different quantities. Neither is a direct count of deaths caused by recent climate change.

03 MOVING INTO THE HEAT

Net domestic migration, July 2023-July 2024

Maricopa County, AZ	-2,049
Pinal County, AZ	+23,316
Phoenix metro total	+21,267

Phoenix recorded 70 days at 110°F or hotter in 2024. The metro still gained domestic movers, but these totals do not reveal where they came from or how their personal exposure changed.

04 PROTECTION PEOPLE CAN USE

My first reaction was that Europe needed more AC. But owning a unit does not mean being able to afford electricity. I would compare targeted cooling and bill support with housing improvements, counting costs and health benefits.

Hold on: a falling death trend does not isolate adaptation, and AC users' better outcomes do not prove a subsidy will deliver the same benefit. Evaluate the program, not just the equipment.

Sources: NWS, long-run fatalities & definitions: weather.gov/hazstat. Census, annual population and Vintage 2025 county migration: census.gov/popest. NWS, 2024 Phoenix climate review: weather.gov/psr/yearinreview2024. Burke et al. (2025), NBER Working Paper 34313: nber.org/papers/w34313. All source links and calculations are on the webpage.